Parameters of B-spline set are $K = 9$, NBF $= 100$, $R = 40$, and $R_1 = 10^{-4}$. Exponential knot sequence is used.

TABLE I: DK3-SCF energies and energy deviations from NH's values ΔE.

Z	A	Atom	Configuration	State	DK3-SCF energies (hartree)		ΔE (mhartree)
					This work	Nakajima and Hirao[a]	
2	4	He	$1s^2$	^{1}S	−2.861781433	−2.861253210	−0.528223
3	7	Li	[He]$2s^1$	^{2}S	−7.433401899	−7.433370002	−0.031897
4	9	Be	[He]$2s^2$	^{1}S	−14.57554039	−14.57548338	−0.05701
5	11	B	[He]$2s^22p^1$	^{2}P	−24.53582114	−24.53572137	−0.09977
6	12	C	[He]$2s^22p^2$	^{3}P	−37.70362775	−37.70346764	−0.16011
7	14	N	[He]$2s^22p^3$	^{4}S	−54.43027432	−54.43003251	−0.24181
8	16	O	[He]$2s^22p^4$	^{3}P	−74.86176921	−74.86139693	−0.37228
9	19	F	[He]$2s^22p^5$	^{2}P	−99.49658346	−99.49604040	−0.54306
10	20	Ne	[He]$2s^22p^6$	^{1}S	−128.6847789	−128.6840551	−0.7238
11	23	Na	[Ne]$3s^1$	^{2}S	−162.0682402	−162.0681010	−0.1392
12	24	Mg	[Ne]$3s^2$	^{1}S	−199.9218634	−199.9216409	−0.2225
13	27	Al	[Ne]$3s^23p^1$	^{2}P	−242.3134680	−242.3129938	−0.4742
14	28	Si	[Ne]$3s^23p^2$	^{3}P	−289.4589038	−289.4583488	−0.5550
15	31	P	[Ne]$3s^23p^3$	^{4}S	−341.5365976	−341.5359065	−0.6911
16	32	S	[Ne]$3s^23p^4$	^{3}P	−398.5895204	−398.5886882	−0.8322
17	35	Cl	[Ne]$3s^23p^5$	^{2}P	−460.8957358	−460.8947817	−0.9541
18	40	Ar	[Ne]$3s^23p^6$	^{1}S	−528.6319988	−528.6308474	−1.1514
19	39	K	[Ar]$4s^1$	^{2}S	−601.4637437	−601.4632019	−0.5418
20	40	Ca	[Ar]$4s^2$	^{1}S	−679.6360500	−679.6358686	−0.1814
21	45	Sc	[Ar]$4s^23d^1$	^{2}D	−763.2912952	−763.2911079	−0.1873
22	48	Ti	[Ar]$4s^23d^2$	^{3}F	−852.7530223	−852.7527989	−0.2234
23	51	V	[Ar]$4s^23d^3$	^{4}F	−948.1495423	−948.1492636	−0.2787
24	52	Cr	[Ar]$4s^13d^5$	^{7}S	−1049.672351	−1049.671965	−0.386
25	55	Mn	[Ar]$4s^23d^5$	^{6}S	−1157.401927	−1157.401506	−0.421
26	56	Fe	[Ar]$4s^23d^6$	^{5}D	−1271.362016	−1271.361472	−0.544
27	59	Co	[Ar]$4s^23d^7$	^{4}F	−1391.899985	−1391.899367	−0.618
28	58	Ni	[Ar]$4s^23d^8$	^{3}F	−1519.124891	−1519.124180	−0.711
29	63	Cu	[Ar]$4s^13d^{10}$	^{2}S	−1653.188648	−1653.187690	−0.958
30	64	Zn	[Ar]$4s^23d^{10}$	^{1}S	−1794.312151	−1794.311192	−0.959
31	69	Ga	[Ar]$4s^23d^{10}4p^1$	^{2}P	−1942.225081	−1942.223759	−1.322
32	74	Ge	[Ar]$4s^23d^{10}4p^2$	^{3}P	−2097.109414	−2097.108132	−1.282
33	75	As	[Ar]$4s^23d^{10}4p^3$	^{4}S	−2259.082441	−2259.080892	−1.549
34	80	Se	[Ar]$4s^23d^{10}4p^4$	^{3}P	−2428.136452	−2428.134832	−1.620
35	79	Br	[Ar]$4s^23d^{10}4p^5$	^{2}P	−2604.492169	−2604.490450[b]	−1.719
36	84	Kr	[Ar]$4s^23d^{10}4p^6$	^{1}S	−2788.268965	−2788.266866	−2.099
37	85	Rb	[Kr]$5s^1$	^{2}S	−2979.147520	−2979.146576	−0.944
38	88	Sr	[Kr]$5s^2$	^{1}S	−3177.350520	−3177.348340	−2.180
39	89	Y	[Kr]$5s^24d^1$	^{2}D	−3382.954001	−3382.951022	−2.979
40	90	Zr	[Kr]$5s^24d^2$	^{3}F	−3596.215825	−3596.212343	−3.482
41	93	Nb	[Kr]$5s^14d^4$	^{6}D	−3817.270074	−3817.267655	−2.419
42	98	Mo	[Kr]$5s^14d^5$	^{7}S	−4046.235920	−4046.233179	−2.741
43	98	Tc	[Kr]$5s^24d^5$	^{6}S	−4283.094133	−4283.089204	−4.929
44	102	Ru	[Kr]$5s^14d^7$	^{5}F	−4528.036801	−4528.033569	−3.232
45	103	Rh	[Kr]$5s^14d^8$	^{4}F	−4781.243759	−4781.240333	−3.426
46	106	Pd	[Kr]$4d^{10}$	^{1}S	−5042.810705	−5042.809114	−1.591
47	107	Ag	[Kr]$5s^14d^{10}$	^{2}S	−5312.889572	−5312.885646	−3.926
48	114	Cd	[Kr]$5s^24d^{10}$	^{1}S	−5591.406237	−5591.404045	−2.192
49	115	In	[Kr]$5s^24d^{10}5p^1$	^{2}P	−5878.336621	−5878.334205	−2.416
50	120	Sn	[Kr]$5s^24d^{10}5p^2$	^{3}P	−6173.852425	−6173.849993	−2.432
51	121	Sb	[Kr]$5s^24d^{10}5p^3$	^{4}S	−6478.069341	−6478.066994	−2.347
52	130	Te	[Kr]$5s^24d^{10}5p^4$	^{3}P	−6790.977565	−6790.975205	−2.360
53	127	I	[Kr]$5s^24d^{10}5p^5$	^{2}P	−7112.797074	−7112.794678	−2.396
54	132	Xe	[Kr]$5s^24d^{10}5p^6$	^{1}S	−7443.622118	−7443.619596	−2.522
55	133	Cs	[Xe]$6s^1$	^{2}S	−7783.199577	−7783.197869	−1.708
56	138	Ba	[Xe]$6s^2$	^{1}S	−8131.748810	−8131.746522	−2.288
57	139	La	[Xe]$6s^25d^1$	^{2}D	−8489.398628	−8489.395578	−3.050
58	140	Ce	[Xe]$6s^24f^15d^1$	^{1}G	−8856.470752	−8856.467882	−2.870
59	141	Pr	[Xe]$6s^24f^3$	^{4}I	−9233.223828	−9233.221418	−2.410
60	144	Nd	[Xe]$6s^24f^4$	^{5}I	−9619.825306	−9619.822739[b]	−2.567
61	145	Pm	[Xe]$6s^24f^5$	^{6}H	−10016.35875	−10016.35607[b]	−2.68
62	152	Sm	[Xe]$6s^24f^6$	^{7}F	−10422.98978	−10422.98719	−2.59
63	153	Eu	[Xe]$6s^24f^7$	^{8}S	−10839.90883	−10839.90614	−2.69
64	158	Gd	[Xe]$6s^24f^75d^1$	^{9}D	−11267.09554	−11267.09204	−3.50
65	159	Tb	[Xe]$6s^24f^9$	^{6}H	−11704.49948	−11704.49650	−2.98
66	162	Dy	[Xe]$6s^24f^{10}$	^{5}I	−12152.74959	−12152.74652[b]	−3.07
67	162	Ho	[Xe]$6s^24f^{11}$	^{4}I	−12611.78717	−12611.78126[b]	−5.91
68	168	Er	[Xe]$6s^24f^{12}$	^{3}H	−13081.72826	−13081.10304	−625.22
69	169	Tm	[Xe]$6s^24f^{13}$	^{2}F	−13562.82458	−13561.50358	−1321.00
70	174	Yb	[Xe]$6s^24f^{14}$	^{1}S	−14055.22080	−14053.57695	−1643.85
71	175	Lu	[Xe]$6s^24f^{14}5d^1$	^{2}D	−14559.03503	−14559.02934	−5.69
72	180	Hf	[Xe]$6s^24f^{14}5d^2$	^{3}F	−15074.18709	−15074.18165	−5.44

TABLE I: DK3-SCF energies and energy deviations from NH's values ΔE.

Z	A	Atom	Configuration	State	DK3-SCF energies (hartree)		ΔE (mhartree)
					This work	Nakajima and Hirao[a]	
73	181	Ta	$[Xe]6s^24f^{14}5d^3$	^{4}F	−15600.84816	−15600.84227	−5.89
74	184	W	$[Xe]6s^24f^{14}5d^4$	^{5}D	−16139.12987	−16139.12306	−6.81
75	187	Re	$[Xe]6s^24f^{14}5d^5$	^{6}S	−16689.22439	−16689.21739	−7.00
76	192	Os	$[Xe]6s^24f^{14}5d^6$	^{5}D	−17251.10270	−17251.09546	−7.24
77	193	Ir	$[Xe]6s^24f^{14}5d^7$	^{4}F	−17825.13361	−17825.12572	−7.89
78	195	Pt	$[Xe]6s^14f^{14}5d^9$	^{3}D	−18411.42855	−18411.42321	−5.34
79	197	Au	$[Xe]6s^14f^{14}5d^{10}$	^{2}S	−19010.19012	−19010.18469	−5.43
80	202	Hg	$[Xe]6s^24f^{14}5d^{10}$	^{1}S	−19621.43452	−19621.43186	−2.66
81	205	Tl	$[Hg]6p^1$	^{2}P	−20245.17239	−20245.16931	−3.08
82	208	Pb	$[Hg]6p^2$	^{3}P	−20881.66356	−20881.66073	−2.83
83	209	Bi	$[Hg]6p^3$	^{4}S	−21531.12028	−21531.11761	−2.67
84	209	Po	$[Hg]6p^4$	^{3}P	−22193.61560	−22193.61322	−2.38
85	210	At	$[Hg]6p^5$	^{2}P	−22869.39895	−22869.39671	−2.24
86	222	Rn	$[Hg]6p^6$	^{1}S	−23558.47172	−23558.46968	−2.04
87	223	Fr	$[Rn]7s^1$	^{2}S	−24261.09851	−24261.09889	0.38
88	226	Ra	$[Rn]7s^2$	^{1}S	−24977.36490	−24977.36461	−0.29
89	227	Ac	$[Rn]7s^26d^1$	^{2}D	−25707.53344	−25707.53428	0.84
90	232	Th	$[Rn]7s^26d^2$	^{3}F	−26451.77385	−26451.77497	1.12
91	231	Pa	$[Rn]7s^25f^26d^1$	^{4}K	−27210.64872	−27210.64973	1.01
92	238	U	$[Rn]7s^25f^36d^1$	^{5}L	−27984.17410	−27984.17530	1.20
93	237	Np	$[Rn]7s^25f^46d^1$	^{6}L	−28773.01975	−28773.02137	1.62
94	244	Pu	$[Rn]7s^25f^6$	^{7}F	−29576.89254	−29576.89439[b]	1.85
95	243	Am	$[Rn]7s^25f^7$	^{8}S	−30396.77948	−30396.78233	2.85
96	247	Cm	$[Rn]7s^25f^76d^1$	^{9}D	−31232.27291	−31232.27592	3.01
97	247	Bk	$[Rn]7s^25f^86d^1$	^{8}H	−32083.97596	−32083.97945	3.49
98	251	Cf	$[Rn]7s^25f^{10}$	^{5}I	−32952.05375	−32952.05808	4.33
99	252	Es	$[Rn]7s^25f^{11}$	^{4}I	−33837.09148	−33837.09652	5.04
100	257	Fm	$[Rn]7s^25f^{12}$	^{3}H	−34738.98956	−34738.99548	5.92
101	258	Md	$[Rn]7s^25f^{13}$	^{2}F	−35658.61813	−35658.62486	6.73
102	259	No	$[Rn]7s^25f^{14}$	^{1}S	−36596.11435	−36596.12201	7.66
103	262	Lr	$[Rn]7s^25f^{14}6d^1$	^{2}D	−37551.42153	−37551.42946	7.93
104	261	Rf	$[Rn]7s^25f^{14}6d^2$	^{3}F	−38525.43721		
105	262	Db	$[Rn]7s^25f^{14}6d^3$	^{4}F	−39518.03195		
106	263	Sg	$[Rn]7s^25f^{14}6d^4$	^{5}D	−40529.77688		
107	262	Bh	$[Rn]7s^25f^{14}6d^5$	^{6}S	−41561.36427		
108	265	Hs	$[Rn]7s^25f^{14}6d^6$	^{5}D	−42612.35520		
109	265	Mt	$[Rn]7s^25f^{14}6d^7$	^{4}F	−43684.08704		
110	271	Ds	$[Rn]7s^15f^{14}6d^9$	^{3}D	−44776.04341		
111	272	Rg	$[Rn]7s^15f^{14}6d^{10}$	^{2}S	−45890.16526		

[a] Ref. 7

[b] The results were recalculated by the present author.

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